Stretching VLAN 40 from DC1 to DC3

How two data centers share one Layer-2 segment through EVPN VXLAN gateways: route types, control plane and packet flow

Item	Detail
What	VLAN 40 (10.40.40.0/24, VNI 110040) exists in DC1 and DC3 as one Layer-2 segment
How	EVPN VXLAN gateway on each border leaf (RFC 9014), one EVPN domain per DC
Test hosts	VPC1 10.40.40.10 (DC1, leaf1) and VPC6 10.40.40.60 (DC3, dc3-leaf1)
Route types involved	**Type-3** (find each other, flood lists) and **Type-2** (exact host location)
Status	Working: VPC1 ↔ VPC6 ping successful

1. The goal in one sentence

A host in DC1 and a host in DC3 sit in **the same subnet (10.40.40.0/24)** and talk to each other **as if they were plugged into the same switch**, while each data center keeps its own independent EVPN fabric.

Why you would do this in production: VM migration (vMotion) between sites, clustered applications that need a shared subnet, or moving a workload to another DC without changing its IP address.

2. The idea in plain language

2.1 Main Street runs through two cities

Picture DC1 and DC3 as two **cities**, each with its own postal system. **VLAN 40 is "Main Street"**, and it runs through both cities. VPC1 lives on Main Street in DC1; VPC6 lives on Main Street in DC3.

Each city has a **customs office** at its border (the border leaf with the EVPN gateway), and a **bridge** connects the two customs offices (the metro DWDM link, Eth5).

Network	Real world
VLAN 40 / VNI 110040	Main Street, one street name in both cities
VPC1, VPC6	Houses on Main Street
dc1-border-leaf, dc3-border-leaf (EVPN gateways)	Customs offices at each city border
Metro link Eth5	The bridge between the customs offices
Type-2 route	A **letter addressed to a named person**: "VPC6 lives at this address"
Type-3 route	The **flyer distribution list**: "who wants copies of Main Street broadcasts"
VXLAN tunnel	A postman carrying a sealed envelope between two post offices
Outer IP header	The address on the envelope (changes at each customs office)
Inner Ethernet frame	The letter inside (never opened, never changed)

2.2 What the customs offices do

- When DC3 says "VPC6 lives here", DC3's customs office **re-stamps** the message: "VPC6 lives behind **me**". DC1's customs office re-stamps it again for DC1: "VPC6 lives behind **me**".
- So a DC1 leaf only ever learns "send VPC6's mail to my own customs office". It never learns DC3's internal addresses. That keeps each city independent: a problem in one city stays in that city.
- A letter from VPC1 to VPC6 is carried by **three postmen** (three VXLAN tunnels): DC1 leaf → DC1 customs → DC3 customs → DC3 leaf. The envelope address changes at each hop; the letter inside never does.

> **Memory hooks**
>
> **Customs re-stamps the letter**: each gateway rewrites the route's next hop (and RD) to itself.
>
> **Three postmen, one envelope**: three tunnels, the inner frame never changes.

Type-3 to find, Type-2 to deliver.
Same VNI everywhere: 110040 in DC1, on the metro, and in DC3.

3. The picture

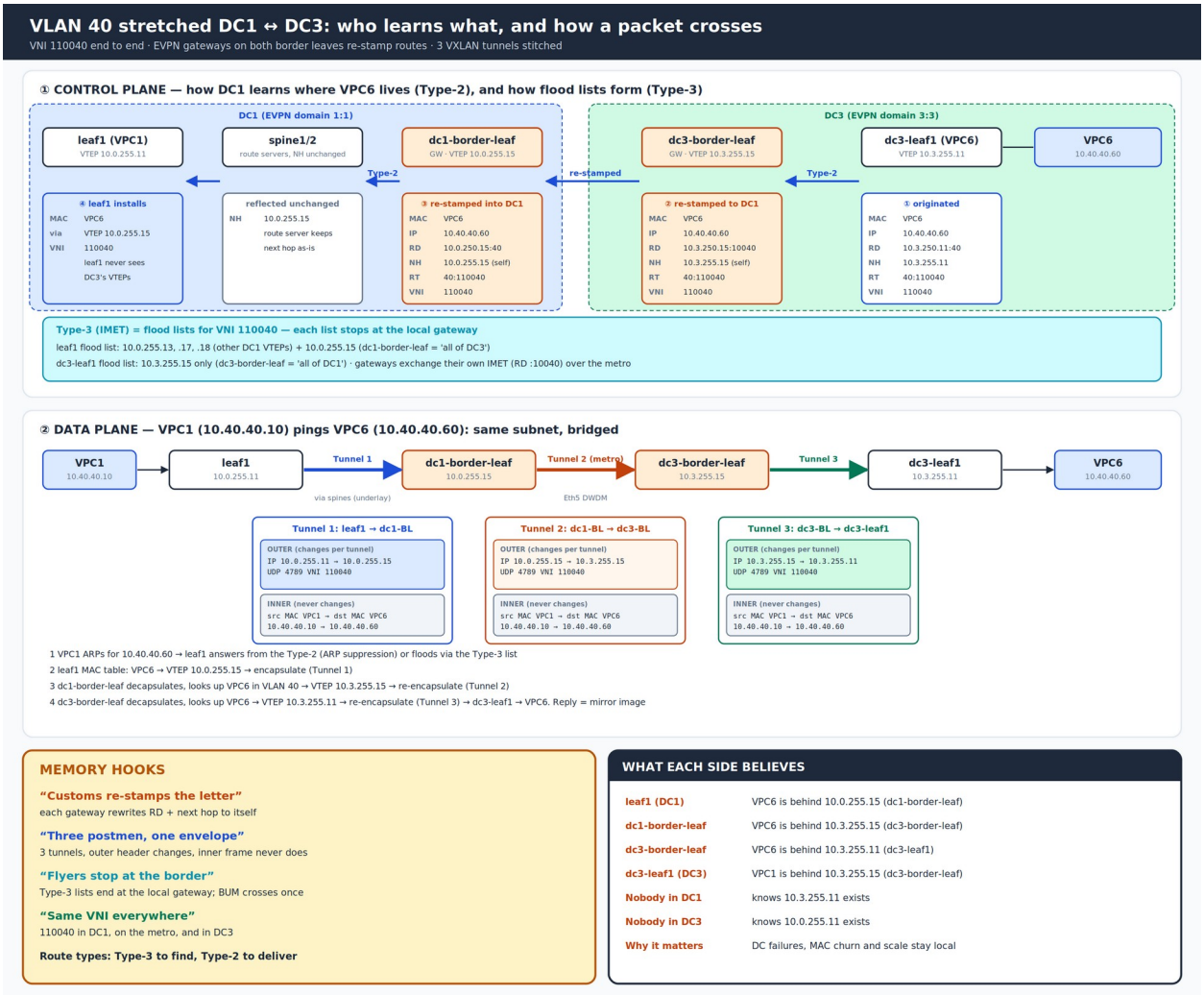


Figure 1: Control plane (how routes are learned and re-stamped) and data plane (how a frame crosses three tunnels)

3.1 The players

Device	Role for VLAN 40	VTEP (Lo1)	EVPN peer(s)
leaf1 (+ leaf2, MLAG)	Where VPC1 lives; anycast GW 10.40.40.1	10.0.255.11 (shared MLAG VTEP)	spine1/2 (route servers)
spine1 / spine2	Relay EVPN routes inside DC1, next hop unchanged	—	All DC1 leaves
dc1-border-leaf	**DC1 EVPN gateway**	10.0.255.15	spines (DC1 domain) + dc3-border-leaf (remote domain)
dc3-border-leaf	**DC3 EVPN gateway**	10.3.255.15	dc3-leaf1 (DC3 domain) + dc1-border-leaf (remote domain)
dc3-leaf1	Where VPC6 lives; anycast GW 10.40.40.1	10.3.255.11	dc3-border-leaf

4. Five ideas you need first

Idea	Plain meaning	In this lab
L2VNI	The VXLAN ID for one Layer-2 segment. Same VNI = same VLAN everywhere	110040 for VLAN 40 in both DCs
MAC-VRF	The EVPN "table" for one VLAN: which MACs live where	`router bgp` → `vlan 40` block
EVPN domain	One fabric that shares EVPN routes freely	DC1 = domain 1:1, DC3 = domain 3:3
Route Distinguisher (RD)	A "who sent this" tag that makes a route unique	Each device and each domain uses its own RD
Route Target (RT)	A "which table does this belong to" tag; matching RT = import	40:110040 everywhere for VLAN 40

> **The key trick: gateway re-origination**
>
> A normal EVPN router passes routes along. A **gateway** (configured with `domain remote`) **re-originates** them: it receives a route from one domain, then advertises a fresh copy into the other domain with **its own RD and its own VTEP as next hop**.
>
> That is why DC1 never needs to reach DC3's leaf VTEPs, and the underlay between DCs only carries the two gateway loopbacks.

5. Control plane: how each side learns about the other

5.1 Step 1: Type-3 builds the flood lists

Before any host talks, every VTEP that has VLAN 40 advertises a **Type-3 (IMET)** route saying "send me VLAN 40 broadcasts". This builds the **flood list** (ingress replication list) used for ARP and other BUM traffic.

Device	Type-3 it advertises	Where it goes
dc3-leaf1	VTEP 10.3.255.11, RD 10.3.250.11:40, VNI 110040	To dc3-border-leaf (DC3 domain)
dc3-border-leaf	Its own VTEP 10.3.255.15, RD 10.3.250.15:40 (into DC3) and RD 10.3.250.15:10040 (to DC1)	Both domains
dc1-border-leaf	Its own VTEP 10.0.255.15, RD 10.0.250.15:40 (into DC1) and RD 10.0.250.15:10040 (to DC3)	Both domains
DC1 leaves	Their own VTEPs, RD Lo0:40	DC1 domain only

VTEP	Its flood list for VNI 110040	Meaning
leaf1	10.0.255.13, 10.0.255.17, 10.0.255.18, **10.0.255.15**	Other DC1 leaves + dc1-border-leaf (which stands for "all of DC3")
dc1-border-leaf	DC1 leaves + **10.3.255.15**	All of DC1 + dc3-border-leaf
dc3-border-leaf	**10.0.255.15** + 10.3.255.11	dc1-border-leaf + dc3-leaf1
dc3-leaf1	**10.3.255.15** only	dc3-border-leaf stands for "all of DC1"

Result: a broadcast crosses the metro link **once**, and each flood list stops at the local gateway ("flyers stop at the border").

5.2 Step 2: Type-2 tells DC1 where VPC6 lives

As soon as VPC6 sends any frame, dc3-leaf1 learns its MAC (and IP from ARP) and advertises a **Type-2 (MAC/IP)** route. Watch how the route changes at each customs office:

Hop	Device	RD	Next hop	What happened
①	dc3-leaf1	10.3.250.11:40	10.3.255.11	**Originated**: VPC6 MAC + 10.40.40.60, RT 40:110040, VNI 110040
②	dc3-border-leaf	10.3.250.15:10040	**10.3.255.15** (self)	**Re-stamped** toward DC1 using its remote-domain RD
③	dc1-border-leaf	10.0.250.15:40	**10.0.255.15** (self)	**Re-stamped** into DC1 using its local RD
④	spine1/2	unchanged	10.0.255.15 (unchanged)	Reflected to all DC1 leaves (route servers keep next hop)
⑤	leaf1	—	10.0.255.15	**Installed**: "VPC6 is behind VTEP 10.0.255.15, VNI 110040"

5.3 The same in reverse: DC3 learns where VPC1 lives

Hop	Device	RD	Next hop
①	leaf1 (MLAG pair)	10.0.250.11:40	10.0.255.11
②	dc1-border-leaf	10.0.250.15:10040	**10.0.255.15** (self)
③	dc3-border-leaf	10.3.250.15:40	**10.3.255.15** (self)
④	dc3-leaf1 installs	—	VPC1 is behind 10.3.255.15

5.4 What each device ends up believing

Device	Belief about the other DC's host
leaf1	VPC6 is behind **10.0.255.15** (dc1-border-leaf)
dc1-border-leaf	VPC6 is behind **10.3.255.15** (dc3-border-leaf)
dc3-border-leaf	VPC6 is behind **10.3.255.11** (dc3-leaf1, local)
dc3-leaf1	VPC1 is behind **10.3.255.15** (dc3-border-leaf)
Nobody in DC1	knows that 10.3.255.11 exists
Nobody in DC3	knows that 10.0.255.11 exists

6. Data plane: VPC1 pings VPC6

6.1 ARP: "who is 10.40.40.60?"

1. VPC1 broadcasts an ARP request in VLAN 40.
2. **If leaf1 already has VPC6's Type-2 (MAC + IP)**, it answers the ARP itself (**ARP suppression**). The broadcast never leaves leaf1.
3. **If not**, leaf1 floods the ARP to every VTEP on its **Type-3 flood list**, including dc1-border-leaf. dc1-border-leaf forwards it over the metro to dc3-border-leaf, which forwards it to dc3-leaf1. VPC6 replies, and its MAC is now learned and advertised as a Type-2.

6.2 The ping, hop by hop

Hop	Device	Lookup	Action	Outer IP (src → dst)	VNI
1	VPC1	ARP cache: VPC6 MAC	Sends frame VPC1-MAC → VPC6-MAC in VLAN 40	—	—
2	leaf1	MAC table VLAN 40: VPC6 → VTEP 10.0.255.15	Encapsulate (Tunnel 1), send via spines	10.0.255.11 → 10.0.255.15	110040
3	spine1/2	Underlay route to 10.0.255.15	Plain IP routing (does not look inside)	unchanged	—
4	dc1-border-leaf	Decap; MAC table VLAN 40: VPC6 → VTEP 10.3.255.15	Re-encapsulate (Tunnel 2) out Eth5	10.0.255.15 → 10.3.255.15	110040
5	dc3-border-leaf	Decap; MAC table: VPC6 → VTEP 10.3.255.11	Re-encapsulate (Tunnel 3) out Eth1	10.3.255.15 → 10.3.255.11	110040
6	dc3-leaf1	Decap; MAC table: VPC6 → Ethernet1	Deliver plain frame to VPC6	—	—

The **reply** (VPC6 → VPC1) is the mirror image: dc3-leaf1 → 10.3.255.15 → 10.0.255.15 → 10.0.255.11 → VPC1.

6.3 What changes and what never changes

Part of the packet	Changes?	Why
Inner Ethernet frame (VPC1-MAC → VPC6-MAC, 10.40.40.10 → 10.40.40.60)	**Never**	It is bridged: same subnet, nobody routes it
Outer IP header	At each gateway	Each tunnel has its own two endpoints
VNI	Never (110040)	Same L2VNI in DC1, metro and DC3
MTU	—	Each VXLAN header adds 50 bytes; links run MTU 9214

> **Why three tunnels instead of one?**
>
> Without gateways, leaf1 would build a tunnel straight to dc3-leaf1 (10.0.255.11 → 10.3.255.11). DC1 would have to learn every DC3 VTEP and every DC3 MAC change, and one big flat fabric would be one big failure domain.
>
> With gateways, each DC only ever sees one remote VTEP (the other gateway). Scale, MAC churn and

outages stay local.

7. The configuration (VLAN 40 pieces only)

7.1 dc3-leaf1: a normal leaf

dc3-leaf1

```
vlan 40
   name APP-40
interface Ethernet1
   description VPC6
   switchport access vlan 40
interface Vlan40
   vrf gold
   ip address virtual 10.40.40.1/24
interface Vxlan1
   vxlan source-interface Loopback1
   vxlan vlan 40 vni 110040
ip virtual-router mac-address 00:1c:73:00:00:99
router bgp 65301
   neighbor 10.3.250.15 remote-as 65300
   neighbor 10.3.250.15 update-source Loopback0
   neighbor 10.3.250.15 ebgp-multihop 3
   neighbor 10.3.250.15 send-community extended
   vlan 40
      rd 10.3.250.11:40
      route-target both 40:110040
      redistribute learned
   address-family evpn
      neighbor 10.3.250.15 activate
```

Line	What it does	Route type
`vxlan vlan 40 vni 110040`	Maps VLAN 40 to the shared L2VNI	Type-2/3
`vlan 40` → `rd 10.3.250.11:40`	Unique tag for routes from this leaf	Type-2/3
`route-target both 40:110040`	Import/export VLAN 40 routes	Type-2/3
`redistribute learned`	Advertise MACs learned on access ports (VPC6)	**Type-2**
`ip address virtual 10.40.40.1/24` + vMAC	Same gateway IP and MAC as DC1: a VM keeps its gateway after moving	—

7.2 dc3-border-leaf: the DC3 EVPN gateway

dc3-border-leaf

```
vlan 40
   name APP-40
interface Vxlan1
   vxlan source-interface Loopback1
   vxlan vlan 40 vni 110040
!
interface Ethernet5
   mtu 9214
   no switchport
   ip address 172.16.230.1/31
ip prefix-list DCI-UNDERLAY-OUT seq 10 permit 10.3.250.15/32
```

```
ip prefix-list DCI-UNDERLAY-OUT seq 20 permit 10.3.255.15/32
route-map DCI-UNDERLAY-OUT permit 10
   match ip address prefix-list DCI-UNDERLAY-OUT
!
router bgp 65300
   neighbor 172.16.230.0 remote-as 65003
   neighbor 172.16.230.0 route-map DCI-UNDERLAY-OUT out
   neighbor 10.0.250.15 remote-as 65003
   neighbor 10.0.250.15 update-source Loopback0
   neighbor 10.0.250.15 ebgp-multihop 3
   neighbor 10.0.250.15 send-community extended
   neighbor 10.0.250.15 bfd
   vlan 40
      rd 10.3.250.15:40
      rd evpn domain remote 10.3.250.15:10040
      route-target both 40:110040
      route-target import export evpn domain remote 40:110040
      redistribute learned
   address-family evpn
      domain identifier 3:3
      domain identifier 1:1 remote
      neighbor 10.0.250.15 activate
      neighbor 10.0.250.15 domain remote
      neighbor 10.3.250.11 activate
   address-family ipv4
      neighbor 172.16.230.0 activate
      network 10.3.250.15/32
      network 10.3.255.15/32
```

Line	What it does	Route type
`neighbor 172.16.230.0 … route-map DCI-UNDERLAY-OUT out`	Underlay over the metro: share **only** the two gateway loopbacks	IPv4 underlay
`neighbor 10.0.250.15 … ebgp-multihop`	EVPN session to the DC1 gateway, loopback to loopback	—
`neighbor 10.0.250.15 domain remote`	**Makes it a gateway**: routes from DC1 are another domain and get re-originated	Type-2/3
`domain identifier 3:3` / `1:1 remote`	Labels the two domains (loop prevention)	All
`rd 10.3.250.15:40`	RD used when re-stamping DC1's routes **into DC3**	Type-2/3
`rd evpn domain remote 10.3.250.15:10040`	RD used when re-stamping DC3's routes **toward DC1**	Type-2/3
`route-target both 40:110040`	VLAN 40 import/export with dc3-leaf1	Type-2/3
`route-target import export evpn domain remote 40:110040`	VLAN 40 import/export with the DC1 gateway	Type-2/3

7.3 dc1-border-leaf: the DC1 EVPN gateway (mirror image)

dc1-border-leaf (VLAN 40 + metro parts)

```
vlan 40
   name APP-40
interface Vxlan1
   vxlan vlan 40 vni 110040
interface Ethernet5
   mtu 9214
   no switchport
   ip address 172.16.230.0/31
router bgp 65003
   neighbor 172.16.230.1 remote-as 65300
   neighbor 172.16.230.1 route-map DCI-UNDERLAY-OUT out
   neighbor 10.3.250.15 remote-as 65300
   neighbor 10.3.250.15 update-source Loopback0
   neighbor 10.3.250.15 ebgp-multihop 3
   neighbor 10.3.250.15 send-community extended
   neighbor 10.3.250.15 bfd
   vlan 40
      rd 10.0.250.15:40
      rd evpn domain remote 10.0.250.15:10040
      route-target both 40:110040
      route-target import export evpn domain remote 40:110040
      redistribute learned
   address-family evpn
      domain identifier 1:1
      domain identifier 3:3 remote
      neighbor evpn activate
      neighbor 10.3.250.15 activate
      neighbor 10.3.250.15 domain remote
```

DC1 leaves and spines needed **no change**: they already had VLAN 40 / VNI 110040 with RT 40:110040, so they simply import the re-stamped routes from dc1-border-leaf.

8. What happens when...

Event	What the network does	Route type
VPC6 (a VM) moves from DC3 to DC1	The DC1 leaf learns the MAC and advertises a new Type-2 with a higher **MAC mobility sequence number**. Every device prefers the higher sequence, so DC1 and DC3 update within a second. The anycast gateway (same IP and MAC in both DCs) means the VM keeps working without re-ARPing.	Type-2 (sequence number)
The metro link fails	BFD drops the EVPN session; the gateways withdraw the re-stamped routes; VLAN 40 splits into two halves (each DC still works locally). Production fix: two links and two gateways per site.	Type-2/3 withdrawn
A broadcast storm in DC3	Floods follow the Type-3 lists into DC1. Production fix: storm-control and MAC limits on the stretched VLAN.	Type-3
You stretch another VLAN (e.g. 50)	Add the VLAN, VNI and `vlan 50` block (with `domain remote` RD/RT) on both gateways and the DC3 leaf. Nothing else changes.	Type-2/3 for VNI 110050

9. Verify and troubleshoot

Step	Command (device)	Healthy result
1 Underlay	`show ip bgp summary` (both gateways)	172.16.230.x Estab, 2 prefixes received
2 Gateway session	`show bgp evpn summary` (both gateways)	Remote peer Estab
3 Flood lists	`show bgp evpn route-type imet vni 110040` (dc3-leaf1)	An IMET from 10.3.255.15 (the gateway)
4 Host routes	`show bgp evpn route-type mac-ip 10.40.40.60 detail` (leaf1)	NH 10.0.255.15, RD 10.0.250.15:40
5 MAC table	`show vxlan address-table vlan 40` (leaf1, dc3-leaf1)	Remote MAC via the local gateway VTEP
6 Data	`ping 10.40.40.60` (VPC1)	Replies

If the ping fails, work down the list

Step 1 or 2 down → underlay or EVPN session problem on the metro (IP, ASN, multihop, loopback reachability).

No IMET from the gateway (step 3) → VLAN 40 / VNI missing on a gateway, or RT mismatch.

No Type-2 for the remote host (step 4) → the host never sent traffic (ping its gateway first), or `domain remote` RD/RT missing on a gateway.

Routes fine but no replies → MTU on the DCI, or a VXLAN data-plane issue on a gateway.

10. Interview Q&A

How do you extend a VLAN between two EVPN data centers?

EVPN VXLAN gateways on the border leaves (RFC 9014). Each DC is its own EVPN domain; the gateways re-originate Type-2 and Type-3 routes for the stretched VNI with themselves as next hop, so traffic is stitched through three tunnels.

Which route types are involved?

Type-3 builds the flood lists for BUM traffic; Type-2 advertises each host's MAC/IP. Type-5 is only needed for routing between subnets.

Why not just one flat EVPN fabric?

A flat fabric makes every VTEP visible to every other one and turns two DCs into one failure and scale domain. Gateways hide internal VTEPs and limit the blast radius.

How does a VM keep working after moving DCs?

MAC mobility sequence numbers in Type-2 make the new location win, and an anycast gateway (same IP and MAC in both DCs) means the VM's default gateway is already local.

What are the risks of stretching L2?

Broadcast storms and loops span both sites; asymmetric routing with stateful firewalls. Mitigate with storm-control, MAC limits, dual gateways with an interconnect ESI, and treating stretched VLANs as exceptions.